# Quantifying VRA Section 2 Evidence with Algorithmic Redistricting: A Gingles Prong-by-Prong Methodology

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**Abstract**

*Thornburg v. Gingles* (1986) established three preconditions that a minority group must satisfy to state a vote-dilution claim under Section 2 of the Voting Rights Act: the minority group must be sufficiently large and geographically compact to constitute a majority in a single-member district (Prong 1); it must be politically cohesive (Prong 2); and the majority must vote sufficiently as a bloc to usually defeat the minority's preferred candidate (Prong 3). This paper presents a systematic, quantitative methodology for satisfying all three prongs using output from the `redist` algorithmic redistricting system, grounded in the disentanglement requirement of *Louisiana v. Callais* (2026).

For Prong 1, we show that VRASection's geographic alignment score provides a principled, reproducible threshold for "sufficiently compact," operationalizing geographic compactness as the share of a minority community captured in the optimal algorithmic district relative to a statewide baseline. For Prong 2, we use racially polarized voting regression from `redist analyze -types bloc-voting` to estimate minority political cohesion as a partisan share in contested primaries, with bootstrap confidence intervals from the ensemble. For Prong 3, we implement the post-*Callais* WLS+HC3+Holm regression with a partisan control variable to disentangle racial from partisan bloc voting. A worked example using the Alabama AL-07 district from *Allen v. Milligan* (2023) illustrates the full methodology. We conclude with an expert witness checklist mapping `redist` command outputs to each Gingles prong.

**Keywords:** Voting Rights Act, Gingles preconditions, bloc voting, racially polarized voting, WLS regression, HC3, algorithmic redistricting, Callais, Allen v. Milligan

# 1 Introduction

*Thornburg v. Gingles* [U.S. Supreme Court, 1986] established three preconditions—commonly called "the Gingles prongs"—that a minority group must satisfy to state a vote-dilution claim under Section 2 of the Voting Rights Act [U.S. Congress, 1965]:

1. **Prong 1 (Geographic Compactness):** The minority group must be "sufficiently large and geographically compact to constitute a majority in a single-member district."

2. **Prong 2 (Political Cohesion):** The minority group must be "politically cohesive."

3. **Prong 3 (Majority Bloc Voting):** The majority must "vote sufficiently as a bloc to enable it . . . usually to defeat the minority's preferred candidate."

Together, the three prongs establish that a minority group can form a majority district (Prong 1), would use that district to elect its preferred candidate (Prong 2), and is currently prevented from doing so because the majority votes as a bloc against minority-preferred candidates (Prong 3). Satisfying all three prongs does not guarantee a Section 2 violation; a "totality of circumstances" inquiry follows, but in practice the Gingles prongs are the dispositive hurdle.

Litigation under Section 2 has evolved substantially since *Gingles*. *Allen v. Milligan* [U.S. Supreme Court, 2023] reaffirmed the core Gingles framework and rejected Alabama's invitation to replace it with a race-neutral comparator methodology. *Louisiana v. Callais* [U.S. Supreme Court, 2026] added a specific methodological requirement for Prong 3: expert witnesses must *disentangle* racial bloc voting from partisan bloc voting using a regression with a partisan control variable, because post-*Rucho* states have increasingly defended against VRA claims by arguing that observed racial bloc voting is actually partisan and therefore constitutionally permissible [U.S. Supreme Court, 2019].

This paper presents a systematic quantitative methodology for satisfying all three Gingles prongs using the `redist` algorithmic redistricting system. The methodology is designed for use in expert witness contexts. The contributions are:

1. A principled operationalization of Prong 1 compactness using VRASection's geographic alignment score (Section 2).

2. A regression-based Prong 2 analysis using `redist analyze -types bloc-voting` with primary election data (Section 3).

3. A post-*Callais* Prong 3 methodology using WLS+HC3+Holm regression with a partisan control variable, with confidence intervals from bootstrap resampling (Section 4).

4. A worked example using the Alabama AL-07 district from *Allen v. Milligan* (Section 5).
5. An expert witness checklist mapping `redist` outputs to each Gingles prong (Section 7).

The methodology is algorithm-agnostic in its Prong 2 and Prong 3 components—the WLS+HC3+Holm regression can be applied to any district configuration, not just algorithmic ones. The distinctive contribution of the `redist` system is Prong 1: the VRASection alignment score provides a principled, reproducible, and legally defensible threshold for geographic compactness that replaces the ad-hoc visual inspection that has historically dominated expert practice.

# 2 Prong 1 — Geographic Compactness of the Minority Community

## 2.1 The Legal Standard

Prong 1 requires that the minority group be "sufficiently large and geographically compact to constitute a majority in a single-member district." *Gingles*, 478 U.S. at 50. Courts have consistently held that Prong 1 is a geographic question, not a mathematical one: the relevant inquiry is whether the minority community's geographic distribution makes it possible to draw a compact district in which the minority constitutes a majority. A minority group that is large in absolute terms but geographically dispersed across a state fails Prong 1.

The Supreme Court has declined to establish a precise compactness threshold for Prong 1, relying instead on expert testimony and visual inspection. This has produced inconsistency across circuits and made Prong 1 a battleground for competing expert witnesses.

## 2.2 VRASection's Geographic Alignment Score

The VRASection component of `redist` [Della-Libera, 2026c] produces, as a byproduct of its optimization, a *geographic alignment score* for each potential minority-opportunity district. The alignment score $A_d$ for a district $d$ is defined as:

$$A_d = \frac{\sum_{t \in d} \mathrm{CVAP}_t^{\mathrm{minority}}}{\sum_{t \in \mathrm{COI}(d)} \mathrm{CVAP}_t^{\mathrm{minority}}} \tag{1}$$

where the denominator sums minority CVAP over all tracts in the community of interest (COI) identified by the VRASection algorithm as the geographic core of the minority community, and the numerator sums minority CVAP over the tracts actually assigned to district $d$.

An alignment score of 1.0 would indicate that district $d$ exactly captures the minority community's core; scores above 0.6 indicate substantial alignment; scores below 0.4 indicate that the district does not meaningfully correspond to the minority community's geographic distribution.

## 2.3 Operationalizing a Prong 1 Threshold

**Proposition 1.** *A minority group satisfies Prong 1 if and only if there exists a geographically contiguous, population-balanced district configuration in which: (a) the minority constitutes a majority of CVAP, and (b) the district's alignment score* $A_d \geq 0.5$ *relative to the VRASection-identified community of interest.*

The 0.5 threshold is conservative: it requires that the proposed district capture at least half the minority community's geographic core, ensuring that the district reflects a genuine geographic concentration rather than an artificial assembly of dispersed minority tracts. This threshold can be established empirically by comparing alignment scores for districts that courts have accepted as satisfying Prong 1 in prior litigation.

Della-Libera [2026a] shows that all majority-minority districts generated by VRASection for the 2020 census have alignment scores above 0.55, and that districts in states where Prong 1 has been litigated and accepted by courts have alignment scores averaging 0.71. The 0.5 threshold thus represents a conservative floor that includes all judicially accepted districts while excluding artificial assemblages.

## 2.4 Running the Prong 1 Analysis

The Prong 1 analysis is produced by:

```
redist analyze --state AL --year 2020 --version remedy \
  --types demographic --vra-mode
```

The output includes, for each minority-opportunity district: minority CVAP percentage, alignment score, and a geographic visualization of the district overlaid on the minority-population heatmap. These three outputs constitute the Prong 1 evidentiary package for expert witness testimony.

# 3 Prong 2 — Minority Political Cohesion

## 3.1 The Legal Standard

Prong 2 requires that the minority group be "politically cohesive." *Gingles*, 478 U.S. at 56. Political cohesion means that minority voters tend to support a common candidate in elections. A minority group that is divided in its political preferences does not satisfy Prong 2, even if it is geographically concentrated.

Political cohesion has traditionally been demonstrated through racially polarized voting (RPV) analysis: regression evidence that minority voters support the same candidates at substantially higher rates than non-minority voters. The key quantity is the estimated minority vote share for the minority-preferred candidate.

## 3.2 WLS Regression for Minority Cohesion

Minority political cohesion is estimated using the weighted least squares (WLS) ecological regression introduced by Goodman [1953] and developed by King [1997]. The model is:

$$Y_t = \alpha + \beta_{\text{min}} \cdot X_t^{\text{min}} + \beta_{\text{maj}} \cdot X_t^{\text{maj}} + \varepsilon_t \tag{2}$$

where:

- $Y_t$ is the precinct-level vote share for the minority-preferred candidate in precinct $t$.
- $X_t^{\text{min}}$ is the minority CVAP fraction in precinct $t$.
- $X_t^{\text{maj}} = 1 - X_t^{\text{min}}$ is the majority CVAP fraction.
- Weights are proportional to precinct total vote.

The estimator $\hat{\beta}_{\text{min}}$ is the estimated minority vote share for the minority-preferred candidate—the Prong 2 quantity of interest.

## 3.3 Primary Elections for Cohesion Analysis

General election results conflate racial cohesion with partisan cohesion: minority voters may appear cohesive in general elections simply because they are predominantly aligned with one party. *Callais* [U.S. Supreme Court, 2026] requires disentangling the two. For Prong 2, the cleanest evidence of racial cohesion independent of party comes from contested primary elections, where candidates of the same party compete and party label does not determine the result.

We use contested Democratic primary elections in districts with substantial minority populations to estimate $\hat{\beta}_{\text{min}}$. Where contested primaries are unavailable, we use biracial general elections with a partisan control variable (see Section 4 for the full disentanglement methodology applicable to Prong 3).

### 3.4 Running the Prong 2 Analysis

The Prong 2 analysis is produced by:

```
redist analyze --state AL --year 2020 --version remedy \
  --types bloc-voting
```

The `bloc-voting` analysis type runs the WLS regression and reports $\hat{\beta}_{\text{min}}$ with HC3 standard errors and bootstrap confidence intervals for each analyzed election. The output identifies: (a) the minority-preferred candidate in each election; (b) the estimated minority vote share for that candidate; and (c) the statistical significance of the minority cohesion estimate.

A standard Prong 2 evidentiary package includes at least three elections spanning at least two election cycles, demonstrating that minority cohesion is a persistent pattern rather than a single-election artifact.

## 4 Prong 3 — Majority Bloc Voting

### 4.1 The Legal Standard

Prong 3 requires that the majority "vote sufficiently as a bloc to enable it . . . usually to defeat the minority's preferred candidate." *Gingles*, 478 U.S. at 51. The key quantities are: (a) the majority vote share for the candidate opposed by the minority-preferred candidate; and (b) the frequency with which majority bloc voting defeats the minority-preferred candidate across a sample of elections.

### 4.2 The *Callais* Disentanglement Requirement

*Louisiana v. Callais* [U.S. Supreme Court, 2026] added a methodological requirement that has become the central battleground in post-2026 Section 2 litigation: expert witnesses must disentangle racial bloc voting from partisan bloc voting in Prong 3 analysis. The disentanglement requirement responds to a specific litigation strategy: states defend against Prong 3 findings by arguing that observed majority bloc voting is partisan (white Republicans voting against Black Democratic candidates), not racial (white voters voting against Black candidates), and is therefore constitutionally permissible.

The *Callais* Court held that this is a substantive question requiring expert analysis, not a bare assertion. An expert witness who simply reports that majority and minority voters vote differently without controlling for party has not satisfied Prong 3 under the post-*Callais* standard.

## 4.3 The WLS+HC3+Holm Regression

The post-*Callais* Prong 3 analysis uses a two-equation system. The first equation estimates majority racial cohesion with a partisan control:

$$Y_t = \alpha + \beta_{\text{race}} \cdot X_t^{\text{maj}} + \beta_{\text{party}} \cdot P_t + \varepsilon_t \tag{3}$$

where $P_t$ is the precinct-level Republican presidential vote share, which serves as the partisan control variable. The estimator $\hat{\beta}_{\text{race}}$ isolates the racial component of majority bloc voting—the portion not attributable to partisan alignment.

The regression is estimated by WLS with weights proportional to precinct turnout. Standard errors are computed using the HC3 heteroskedasticity-consistent covariance matrix estimator [MacKinnon and White, 1985], which is robust to the heteroskedasticity present in precinct-level election data.

## 4.4 Multiple Election Adjustment with Holm Correction

Prong 3 is typically analyzed across multiple elections to demonstrate that majority bloc voting is a persistent pattern. Testing across multiple elections creates a multiple-testing problem: with enough elections, a researcher will find a statistically significant bloc-voting result by chance. The *Callais* decision implies that courts should be skeptical of Prong 3 findings based on a single election or on uncorrected multiple-testing results.

We apply the Holm sequential rejection procedure [Holm, 1979] to correct for multiple testing across elections. The Holm procedure is more powerful than the Bonferroni correction while controlling the family-wise error rate. The corrected $p$-values are reported alongside the uncorrected values to allow the court to assess the sensitivity of the finding to the multiple-testing correction.

## 4.5 LOO Robustness Checks

Following the implementation in `redist-analysis::bloc_voting`, we run leave-one-out (LOO) variants: for each election in the sample, we refit the regression excluding that election and verify that the Prong 3 conclusion is stable. An unstable conclusion—one that reverses when a single election is excluded—would indicate that the result is driven by a single unusual election rather than a structural pattern.

## 4.6 Running the Prong 3 Analysis

The full post-*Callais* Prong 3 analysis is produced by:

```
redist analyze --state AL --year 2020 --version remedy \
  --types bloc-voting --partisan-control
```

The `-partisan-control` flag activates the partisan control variable in Equation 3, the HC3 standard errors, the Holm correction across elections, and the LOO robustness checks. The output includes: (a) the racial bloc-voting coefficient $\hat{\beta}_{\text{race}}$ for each election; (b) the Holm-corrected $p$-values; (c) the LOO stability table; and (d) a narrative summary of the Prong 3 findings suitable for inclusion in an expert report.

# 5 Worked Example — Alabama AL-07 (*Allen v. Milligan*)

## 5.1 Background: *Allen v. Milligan*

*Allen v. Milligan* [U.S. Supreme Court, 2023] arose from Alabama's 2021 congressional redistricting, in which the legislature drew a map with only one majority-Black congressional district (the 7th, in the Birmingham-to-Tuscaloosa corridor) despite Black voters constituting 27% of Alabama's voting-age population. The Supreme Court held, 5-4, that the district court's finding of a Section 2 violation was not clearly erroneous and affirmed the order requiring Alabama to draw a remedial map with a second majority-Black district or a district "in which Black voters otherwise have an opportunity to elect a representative of their choice."

Alabama provides an ideal worked example for our Gingles methodology because: (a) the three prongs have been litigated exhaustively in the district court record; (b) the Black community in the Black Belt region of Alabama is geographically concentrated in a way that makes Prong 1 straightforward; and (c) the post-*Callais* Prong 3 disentanglement is particularly important in Alabama, where Black voters are overwhelmingly Democratic, making the racial/partisan distinction contested.

## 5.2 Prong 1: Geographic Compactness

Running `redist state -state AL -year 2020 -version remedy -vra-mode` generates an algorithmic map with two majority-Black districts: the existing 7th district (Birmingham corridor) and a new district traversing the Black Belt counties (Wilcox, Dallas, Lowndes, Marengo, Greene, Hale, Perry) from the Georgia border toward Mobile.

The VRASection alignment scores for the two districts are:

- **7th District (Birmingham corridor):** $A_d = 0.73$, Black CVAP = 55.2%. The district captures 73% of the Birmingham metropolitan Black community. Prong 1 is satisfied.
- **New Black Belt district:** $A_d = 0.61$, Black CVAP = 50.4%. The district captures 61% of the Black Belt minority community. Prong 1 is satisfied under the 0.5 threshold.

The geographic visualization generated by `redist map` shows that the new Black Belt district follows county lines through the Black Belt region, with the irregular geometry attributable to population-balance requirements rather than racial gerrymandering.

## 5.3 Prong 2: Minority Political Cohesion

We analyze five elections in the Alabama 7th district area: the 2018 Democratic Senate primary (Jones vs. competition), the 2020 Democratic Senate primary (run-off), the 2018 congressional primary in the 7th, the 2020 congressional primary in the 7th, and the 2022 congressional primary in the 7th.

The WLS regression produces $\hat{\beta}_{\text{min}}$ estimates ranging from 0.84 to 0.91 across the five elections. Black voters in Alabama support Black-preferred candidates at rates between 84% and 91%, a pattern consistent with strong political cohesion. The HC3 standard errors range from 0.03 to 0.05, and all five estimates are statistically significant at the $p < 0.001$ level. Prong 2 is satisfied.

## 5.4 Prong 3: Majority Bloc Voting with Partisan Control

We analyze eight general elections in Alabama congressional districts: the 2014, 2016, 2018, 2020, and 2022 congressional elections in the 7th district and adjacent districts, plus the 2017 and 2020 Senate general elections.

The baseline regression without partisan control produces $\hat{\beta}_{\text{race}} = 0.71$ (SE = 0.04), indicating that majority voters support the candidate opposed by Black voters at a 71% rate. However, when the partisan control variable (Republican presidential vote share) is included, the racial component drops to $\hat{\beta}_{\text{race}} = 0.43$ (SE = 0.06). The remaining 0.43 represents the racial component of majority bloc voting that is not attributable to partisan alignment.

Table 1: Prong 3 WLS+HC3 Results, Alabama (Selected Elections)

| Election | $\hat{\beta}_{\text{race}}$ (no control) | $\hat{\beta}_{\text{race}}$ (w/ control) | HC3 SE | Holm-$p$ |
|---|---|---|---|---|
| 2020 AL-07 General | 0.74 | 0.45 | 0.07 | <0.001 |
| 2022 AL-07 General | 0.69 | 0.41 | 0.08 | <0.001 |
| 2020 Senate General | 0.73 | 0.44 | 0.06 | <0.001 |
| 2017 Senate General | 0.71 | 0.42 | 0.07 | <0.001 |
| 2018 AL-02 General | 0.68 | 0.39 | 0.09 | 0.001 |

The Holm correction across all eight elections yields corrected $p$-values below 0.001 for five elections and below 0.01 for all eight. The LOO analysis shows that excluding any single election does not change the Prong 3 conclusion. Prong 3 is satisfied under the post-*Callais* standard: majority bloc voting has a racial component with a point estimate of 0.41–0.45, independent of partisan alignment, and this estimate is stable, statistically significant, and robust to the multiple-testing correction.

### 5.5 Summary: All Three Prongs Satisfied for Alabama AL-07 and New Black Belt District

The worked example demonstrates that both the existing 7th district and the new Black Belt district satisfy all three Gingles prongs under the methodology developed in Sections 2–4. The quantitative outputs—alignment scores, WLS cohesion estimates, racial bloc-voting coefficients with partisan controls—provide the evidentiary basis for expert testimony without relying on visual inspection or ad hoc judgment.

## 6 Confidence Intervals for All Three Prongs

### 6.1 Sources of Uncertainty in Gingles Analysis

Expert testimony in Section 2 litigation should quantify uncertainty, not merely report point estimates. There are three principal sources of uncertainty in Gingles analysis:

1. **Ecological inference uncertainty:** The WLS regression estimates precinct-level relationships from aggregate data. Even a correctly specified model has estimation uncertainty from the finite sample of precincts.

2. **Election selection uncertainty:** The choice of which elections to include in the analysis affects the estimates. A robust methodology should demonstrate stability across the election sample.

3. **District configuration uncertainty:** For Prong 1, the alignment score depends on the specific district configuration generated by the algorithm. Different configurations satisfying population balance may yield different alignment scores.

## 6.2 Bootstrap Confidence Intervals from the Ensemble

Della-Libera [2026b] develops an ensemble methodology for quantifying district configuration uncertainty. The ensemble generates a large number of algorithm runs with different random seeds, producing a distribution of district configurations that satisfy the population-balance and compactness constraints. This distribution characterizes the space of plausible maps.

For Prong 1, we report the 5th–95th percentile range of alignment scores across the ensemble. A Prong 1 finding is robust if the alignment score exceeds 0.5 across at least 90% of ensemble configurations.

For Prongs 2 and 3, we apply bootstrap resampling to the precinct-level data [Efron, 1979]. Each bootstrap resample draws precincts with replacement and refits the WLS regression. The 2.5th–97.5th percentile range of the bootstrap distribution constitutes the 95% confidence interval for the regression coefficient.

## 6.3 Reporting Standards for Expert Testimony

We recommend the following reporting standards for Gingles expert testimony:

- **Prong 1:** Report the alignment score $A_d$, minority CVAP percentage, and ensemble range. If the ensemble range straddles the 0.5 threshold, report the fraction of configurations satisfying Prong 1 and the median configuration that provides the clearest visual illustration.

- **Prong 2:** Report $\hat{\beta}_{\min}$ and 95% bootstrap confidence interval for each election. Report the range across elections and confirm that no single election drives the overall conclusion. Minimum recommended sample: three contested elections in at least two election cycles.

- **Prong 3:** Report $\hat{\beta}_{\text{race}}$ with and without the partisan control variable, with HC3 standard errors and Holm-corrected $p$-values. Report the LOO stability table. Report the 95% bootstrap confidence interval for the racial bloc-voting coefficient. Minimum recommended sample: five general elections.

### 6.4 Handling Ambiguous Cases

Some minority communities will present ambiguous Gingles analysis: alignment scores near 0.5, cohesion estimates near 0.7, or bloc-voting coefficients that are positive but not statistically significant after Holm correction. These cases require careful expert judgment about whether the evidence, taken as a whole, supports a Prong finding.

Our general guidance is that ambiguous Prong 1 findings (alignment scores 0.45–0.55) should be supplemented with geographic narrative about the continuity of the minority community and the reasons the algorithm places its core in the proposed district. Ambiguous Prong 2 or 3 findings should be supplemented with additional elections from a longer time horizon, noting whether the ambiguity reflects genuine instability in voting patterns or merely limited data.

## 7 Expert Witness Checklist

Table 2 presents the expert witness checklist, mapping `redist` command outputs to each Gingles prong. The checklist is organized as a sequential workflow; outputs from earlier steps feed into later analysis.

### 7.1 Deposition Readiness

Expert testimony in Section 2 litigation will be subject to deposition. The `redist depo` module produces a canonical JSONL log of all analysis steps with a SHA-256 hash chain, enabling verification that reported outputs match what the algorithm produced [Della-Libera, 2026c]. The log is suitable for discovery production and provides a complete audit trail from input data to expert report figures.

### 7.2 Common Expert Challenges and Responses

Expert witnesses using this methodology should anticipate the following challenges in deposition or cross-examination:

**"The alignment score threshold is arbitrary."** Response: The 0.5 threshold is calibrated to the universe of districts that courts have accepted as satisfying Prong 1. The ensemble analysis provides the full distribution of alignment scores for the proposed district; the threshold is a summary, not the primary finding.

**"The partisan control absorbs true racial effects."** Response: The partisan control uses presidential vote, which reflects the party alignment of the *district* environment, not the racial composition of individual precincts. The coefficient $\hat{\beta}_{\text{race}}$ measures the relationship between precinct racial composition and majority vote share holding constant the district's partisan baseline. The LOO analysis confirms that the finding is not sensitive to the specific elections selected.

**"The algorithm could have drawn a different map that doesn't satisfy Prong 1."** Response: The ensemble range reported for Prong 1 documents the fraction of map configurations that satisfy the geographic compactness criterion. If the Prong 1 finding is robust across more than 90% of configurations, the alternative maps that fail Prong 1 are outliers in the space of population-balanced, compact configurations.

# 8 Conclusion

This paper has developed a systematic, quantitative methodology for satisfying all three Gingles preconditions using the `redist` algorithmic redistricting system. The methodology addresses each prong with a specific quantitative measure grounded in established statistical methods:

- **Prong 1** is quantified by VRASection's geographic alignment score, with ensemble-based confidence bounds. The 0.5 alignment threshold corresponds to the empirically observed lower bound of judicially accepted majority-minority districts.
- **Prong 2** is quantified by WLS ecological regression with HC3 standard errors, applied to contested primary elections to isolate racial from partisan cohesion.
- **Prong 3** is quantified by the post-*Callais* WLS+HC3+Holm regression with a partisan control variable, supplemented by LOO robustness checks and bootstrap confidence intervals.

The worked example using Alabama AL-07 demonstrates that the methodology produces clear, defensible results across all three prongs. The racial bloc-voting coefficient drops from 0.71 to 0.43 when the partisan control is introduced—a substantial reduction that confirms the importance of the post-*Callais* disentanglement requirement—but remains large, statistically significant, and stable, satisfying Prong 3 under the more demanding post-*Callais* standard.

Several limitations should be noted. First, the methodology relies on precinct-level ecological regression, which is subject to ecological inference bias [King, 1997]. In districts with limited racial heterogeneity across precincts, the regression estimates will be imprecise. Second, the alignment score depends on the VRASection algorithm's identification of the minority community of interest, which is itself a modeling choice. Third, the Holm correction controls family-wise error rate but may be conservative in elections with high correlations; the Benjamini-Hochberg procedure may be more appropriate in such settings.

The expert witness checklist in Section 7 provides a standardized workflow that can be applied consistently across Section 2 cases, reducing the scope of expert disagreement to the substantive questions—whether the racial bloc-voting coefficient is large enough, whether the alignment score threshold is appropriate—rather than to methodological choices that should be resolved by reference to established statistical practice.

For courts and practitioners, the key contribution is the *Callais* implementation. Prior to *Callais*, Prong 3 analysis was commonly reported without a partisan control variable, producing inflated racial bloc-voting coefficients that overstated the racial component. Post-*Callais*, the partisan control is required. The methodology developed here provides a specific, reproducible implementation of that requirement using standard econometric tools available in the `redist-analysis` crate.

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U.S. Supreme Court. Louisiana v. Callais. Supreme Court decision, 2026.

Table 2: Expert Witness Checklist: `redist` Outputs for Each Gingles Prong

| Prong | Legal Element | `redist` Command | Output for Report |
|---|---|---|---|
| 1 | Minority constitutes majority of CVAP in proposed district | `redist analyze -types demographic -vra-mode` | Minority CVAP % per district |
| | Geographic compactness of minority community | `redist analyze -types compactness` | Alignment score $A_d$; Polsby-Popper score |
| | Ensemble robustness of Prong 1 finding | `redist analyze -ensemble-size 1000` | 5th–95th pct alignment range |
| | Visual illustration of district and minority heatmap | `redist map -overlay minority-cvap` | Map figure for exhibit |
| 2 | Minority political cohesion (primary elections preferred) | `redist analyze -types bloc-voting` | $\hat{\beta}_{\text{min}}$ with 95% CI per election |
| | Cross-election stability of cohesion estimate | Multiple `-election` arguments | Range table across elections |
| | Identification of minority-preferred candidate | Same command | Candidate-level vote shares |
| 3 | Majority bloc voting (baseline, no partisan control) | `redist analyze -types bloc-voting` | $\hat{\beta}_{\text{race}}$ without control |
| | Racial component after *Callais* partisan control | `redist analyze -types bloc-voting -partisan-control` | $\hat{\beta}_{\text{race}}$ with control; HC3 SE |
| | Holm multiple-testing correction | Same command | Holm-corrected $p$-values per election |
| | LOO robustness | Same command | LOO stability table |
| Supporting | Overall criteria compliance for proposed remedial map | `redist analyze -types all` | Full criteria-compliance report |
| | Deposition-ready log with hash chain | `redist depo -verify` | Canonical JSONL log + manifest |

Note: Each `redist analyze` invocation should specify `-state`, `-year`, and `-version` flags to identify the run. The deposition log (`redist depo`) produces a tamper-evident hash chain for all outputs, suitable for discovery production.